OXFORD AQA

INTERNATIONAL QUALIFICATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname ______________________________

Forename(s) ______________________________

Candidate signature ______________________________
I declare this is my own work.

INTERNATIONAL GCSE
BIOLOGY
Paper 2

Wednesday 1 November 2023   07:00 GMT   Time allowed: 1 hour 30 minutes

Materials
For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice
In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

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Answer **all** questions in the spaces provided.

0 1 This question is about cells.

0 1 . 1 Where in a cell do most chemical reactions happen?

[1 mark]

Tick (✓) **one** box.

Cytoplasm ☐

Mitochondria ☐

Nucleus ☐

0 1 . 2 A bacterial cell is prokaryotic.

An animal cell is eukaryotic.

Which statement is correct for eukaryotic cells, but **not** for prokaryotic cells?

[1 mark]

Tick (✓) **one** box.

They cause disease ☐

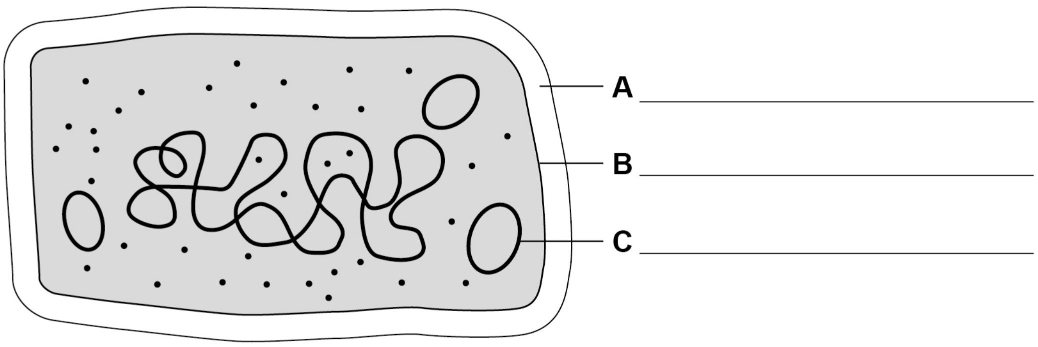
They do not have cytoplasm ☐

They have a nucleus ☐

Do not write outside the box

0 1 . 3 **Figure 1** shows a bacterial cell.

Figure 1

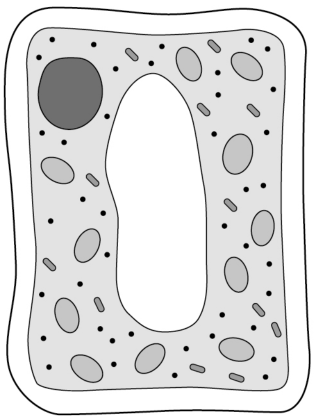


Label structures **A**, **B** and **C** on **Figure 1**.

[3 marks]

0 1 . 4 **Figure 2** shows a plant cell.

Figure 2



Give **two** features in **Figure 2** that show it is a plant cell.

[2 marks]

1 ____

____

2 ____

____

Do not write outside the box

0 1 . 5 A bacterial cell has a width of 4.6 micrometres.

A plant cell has a width of 131.9 micrometres.

Calculate how many times wider the plant cell is than the bacterial cell.

Give your answer to 3 significant figures.

[3 marks]

Number of times wider (3 significant figures) = ____________________

10

Do not write outside the box

0 2 A scientist investigated the effectiveness of new disinfectants at killing bacteria.

This is the method used.

1. Set up a tube of live bacteria in 5 cm^3 of liquid culture medium.
2. Count the number of live bacteria per mm^3 using a microscope.
3. Add 5 cm^3 of disinfectant, **M**.
4. After 12 hours count the number of live bacteria per mm^3 that have survived.
5. Calculate the percentage of live bacteria that have survived.
6. Repeat steps 1 to 5 for disinfectants **N**, **O** and **P**.

0 2 . 1 The scientist carried out a trial before the investigation using a known disinfectant.

In the trial there were:

- 16 000 live bacteria at the start
- 6400 live bacteria after 12 hours.

Calculate the percentage of bacteria that survived.

[2 marks]

Percentage of bacteria that survived = ____________ %

Question 2 continues on the next page

Table 1 shows the results for disinfectants **M**, **N**, **O** and **P**.

Table 1

Disinfectant	**Percentage (%) of bacteria that survived**
M	55
N	25
O	30
P	65

0 2 . 2 Complete **Figure 3**, on the opposite page.

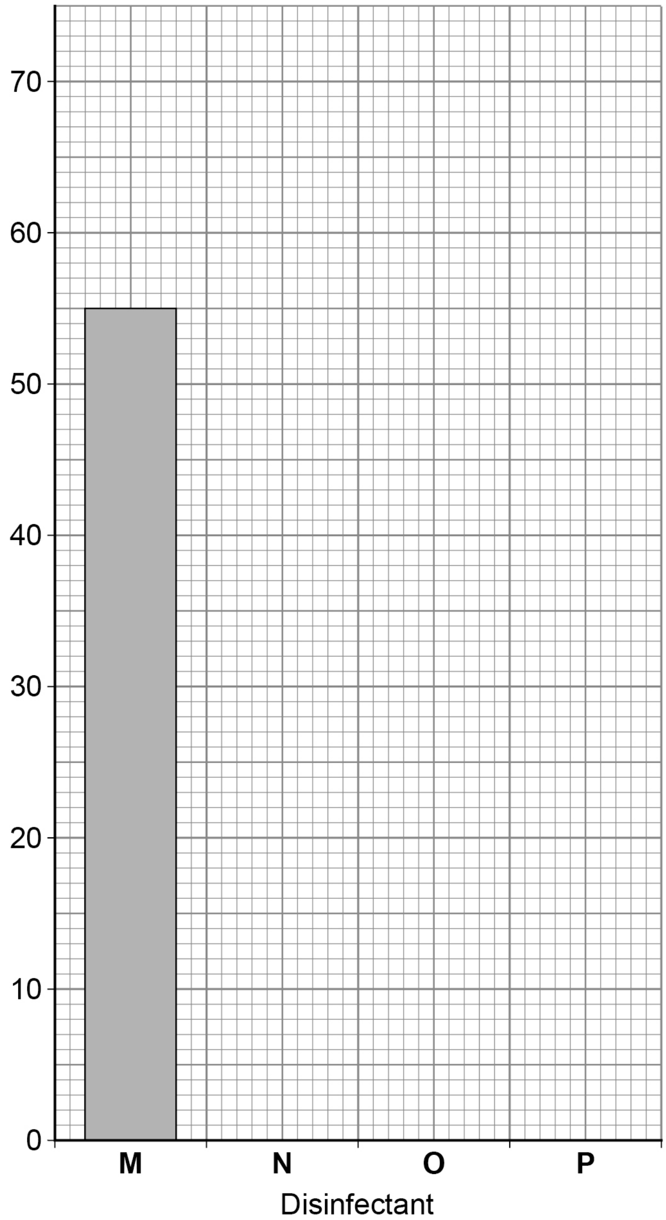
You should:

- label the *y*-axis
- plot the data from **Table 1**.

Disinfectant **M** has been completed for you.

[3 marks]

Figure 3



0 2 . 3 Which disinfectant was the most effective at killing bacteria?

[1 mark]

Tick (✓) **one** box.

M ☐ **N** ☐ **O** ☐ **P** ☐

Do not write outside the box

0 2 . 4 The scientist did not want to contaminate the tubes of bacteria.

Describe **two** techniques the scientist should have used to avoid contamination.

[2 marks]

1 ____

____

2 ____

____

0 2 . 5 Which piece of apparatus would be the most accurate to measure 5 cm^3 of liquid culture medium?

[1 mark]

Tick (✓) **one** box.

10 cm^3 pipette ☐

50 cm^3 beaker ☐

200 cm^3 measuring cylinder ☐

Do not write outside the box

0 2 . 6 The scientist looked at the bacteria through a microscope.

A bacterium has a real size of 10 micrometres.

The image size through the microscope was 2 millimetres.

Calculate the magnification.

Use the equation:

$$\text{magnification} = \frac{\text{image size}}{\text{real size}}$$

1 millimetre = 1000 micrometres

[3 marks]

Magnification = × ____________

12

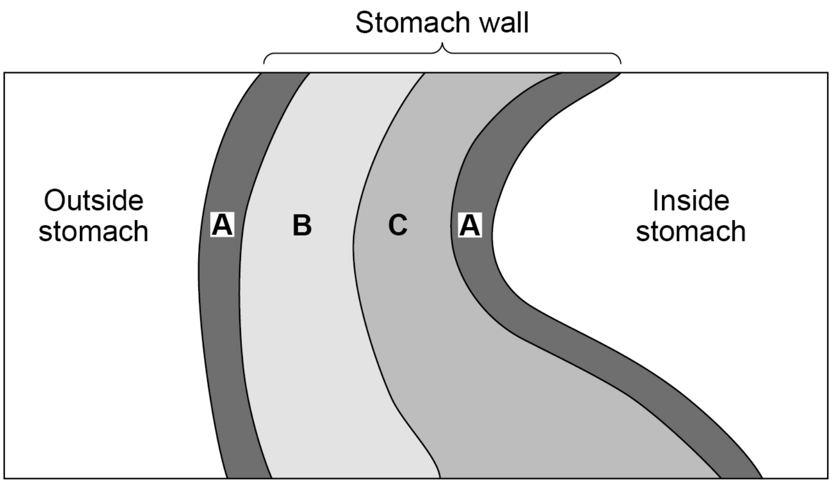
Turn over for the next question

Do not write outside the box

0 3 Organs are made of tissues.

Figure 4 is a simplified diagram of the tissue layers in the stomach wall.

Figure 4



0 3 . 1 Tissue **C** produces digestive enzymes.

Draw **one** line from each tissue to the name of the tissue.

[2 marks]

Tissue	**Name of the tissue**
A	Epithelial tissue
B	Glandular tissue
C	Muscular tissue

0 3 . 2 Describe the function of epithelial tissue and of muscular tissue in the stomach wall.

[2 marks]

Epithelial tissue ____

____

Muscular tissue ____

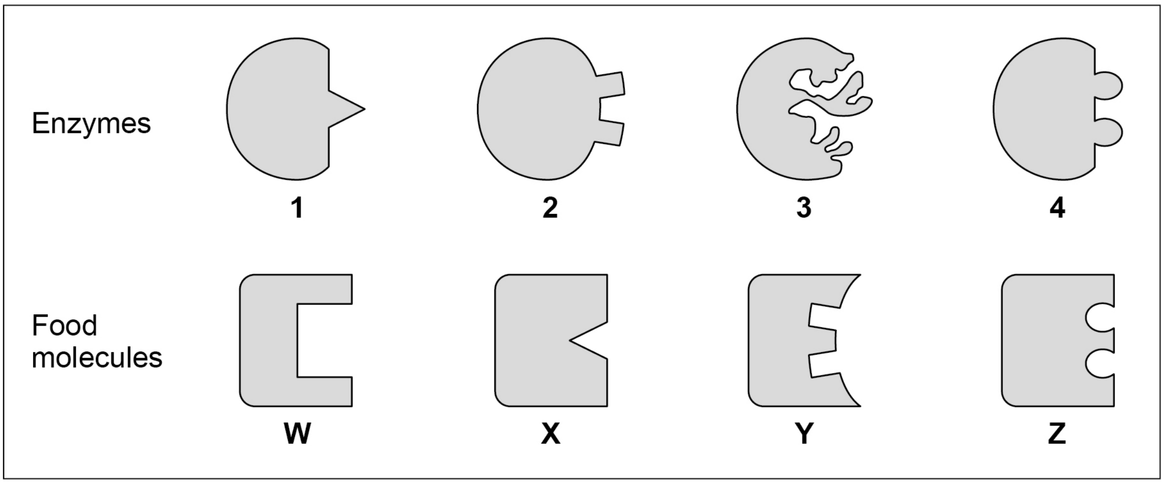
____

Do not write outside the box

Figure 5 shows four digestive enzymes, **1**, **2**, **3** and **4**.

Figure 5 also shows the four food molecules that the enzymes digest, **W**, **X**, **Y**, and **Z**.

Figure 5



0 3 . 3 Which enzyme will break down food molecule **X**?

[1 mark]

Tick (✓) **one** box.

1 ☐ **2** ☐ **3** ☐ **4** ☐

0 3 . 4 Which enzyme has been denatured by heating?

[1 mark]

Tick (✓) **one** box.

1 ☐ **2** ☐ **3** ☐ **4** ☐

0 3 . 5 Why will the denatured enzyme **not** digest any of the food molecules in **Figure 5**?

[1 mark]

____

____

Turn over ►

Do not write outside the box

The large intestine and the liver are two other organs in the digestive system.

0 3 . 6 Describe the function of the large intestine.

[1 mark]

0 3 . 7 Describe the function of the liver in digesting food.

[1 mark]

0 3 . 8 The bile duct connects the gall bladder to the small intestine.

In some people this duct can be blocked by gall stones.

Suggest what effect a blocked bile duct will have on digestion.

[1 mark]

10

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0 4 This question is about photosynthesis.

0 4 . 1 Which equation is the symbol equation for photosynthesis?

[1 mark]

Tick (✓) **one** box.

$6H_2O + 6CO_2 \rightarrow C_6H_{12}O_6 + 6O_2$ ☐

$6H_2O + 6O_2 \rightarrow C_6H_{12}O_6 + 6CO_2$ ☐

$C_6H_{12}O_6 + 6O_2 \rightarrow 6H_2O + 6CO_2$ ☐

$6H_2O + C_6H_{12}O_6 \rightarrow 6CO_2 + 6O_2$ ☐

0 4 . 2 Give **two** ways plants use the glucose made in photosynthesis.

[2 marks]

1 ____

____

2 ____

____

Question 4 continues on the next page

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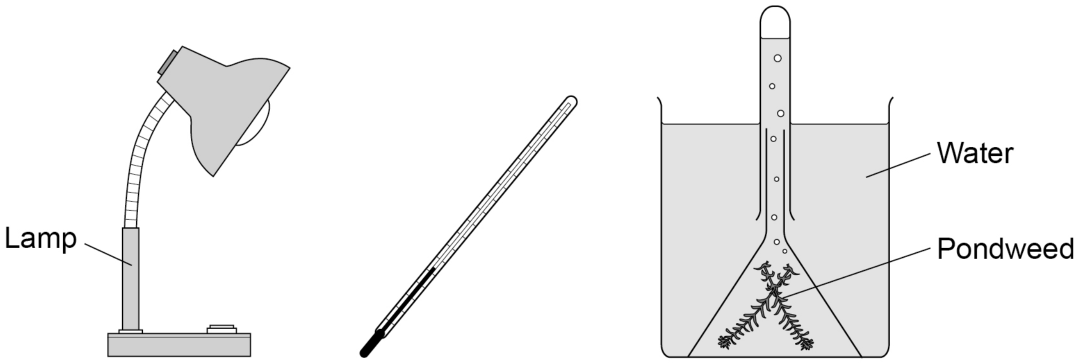
0 4 . 3 Temperature affects the rate of photosynthesis.

Figure 6 shows some of the equipment used to measure the rate of photosynthesis.

Figure 6



Describe a method to investigate the effect of temperature on the **rate** of photosynthesis.

[6 marks]

Do not write outside the box

0 4 . 4 What is the independent variable for the investigation?

[1 mark]

Another student looked at the effect of light intensity on the rate of photosynthesis.

Table 2 shows the results.

Table 2

Light intensity in lux	**Rate of photosynthesis in arbitrary units**		
	Test 1	**Test 2**	**Test 3**
50	15	17	15
100	25	62	27
150	58	55	57
200	63	65	68

0 4 . 5 Draw a ring around the anomalous result.

[1 mark]

0 4 . 6 What should the student do with the anomalous result?

[1 mark]

12

0 5 The core temperature of the body is controlled by homeostasis.

0 5 . 1 Name the part of the brain that coordinates the control of body temperature.

[1 mark]

A scientist investigated the loss of thermal energy from the body.

The scientist:

- measured the thermal energy lost from four different parts of the body
- calculated the thermal energy lost from **each** of the four parts as a percentage of the total thermal energy lost.

Table 3 shows the results.

Table 3

Part of the body	**Percentage (%) of total thermal energy lost from the body**
Arms	15
Torso	30
Head and neck	45
Legs	**X**

0 5 . 2 Value **X** is the percentage of total thermal energy lost from the body by the legs.

Calculate value **X**.

[2 marks]

Value **X** = ________________ %

Do not write outside the box

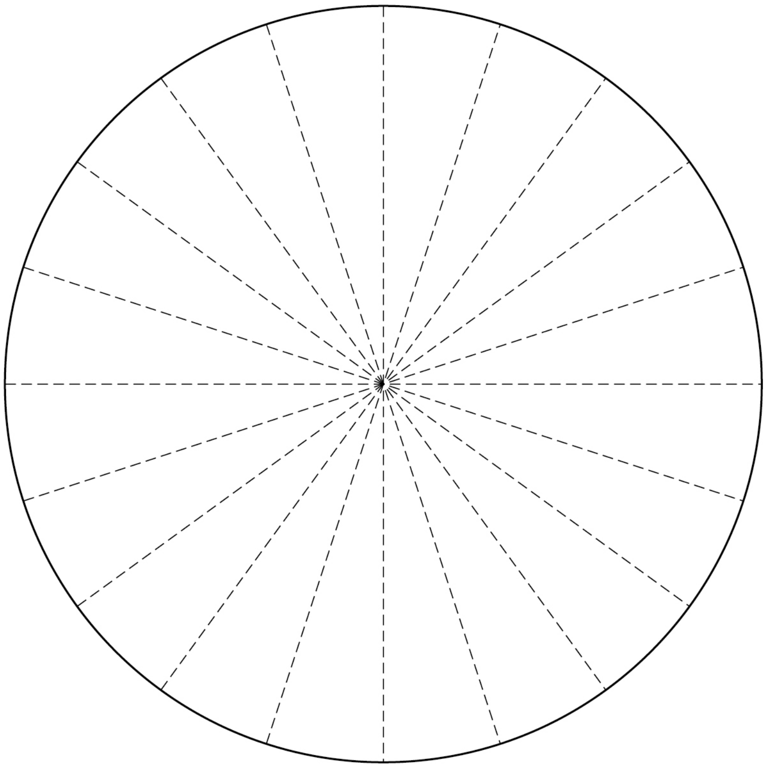
0 5 . 3 Complete the pie chart in **Figure 7**.

Use **Table 3** and your answer to Question **05.2**.

Label the sections of the pie chart.

[2 marks]

Figure 7



Question 5 continues on the next page

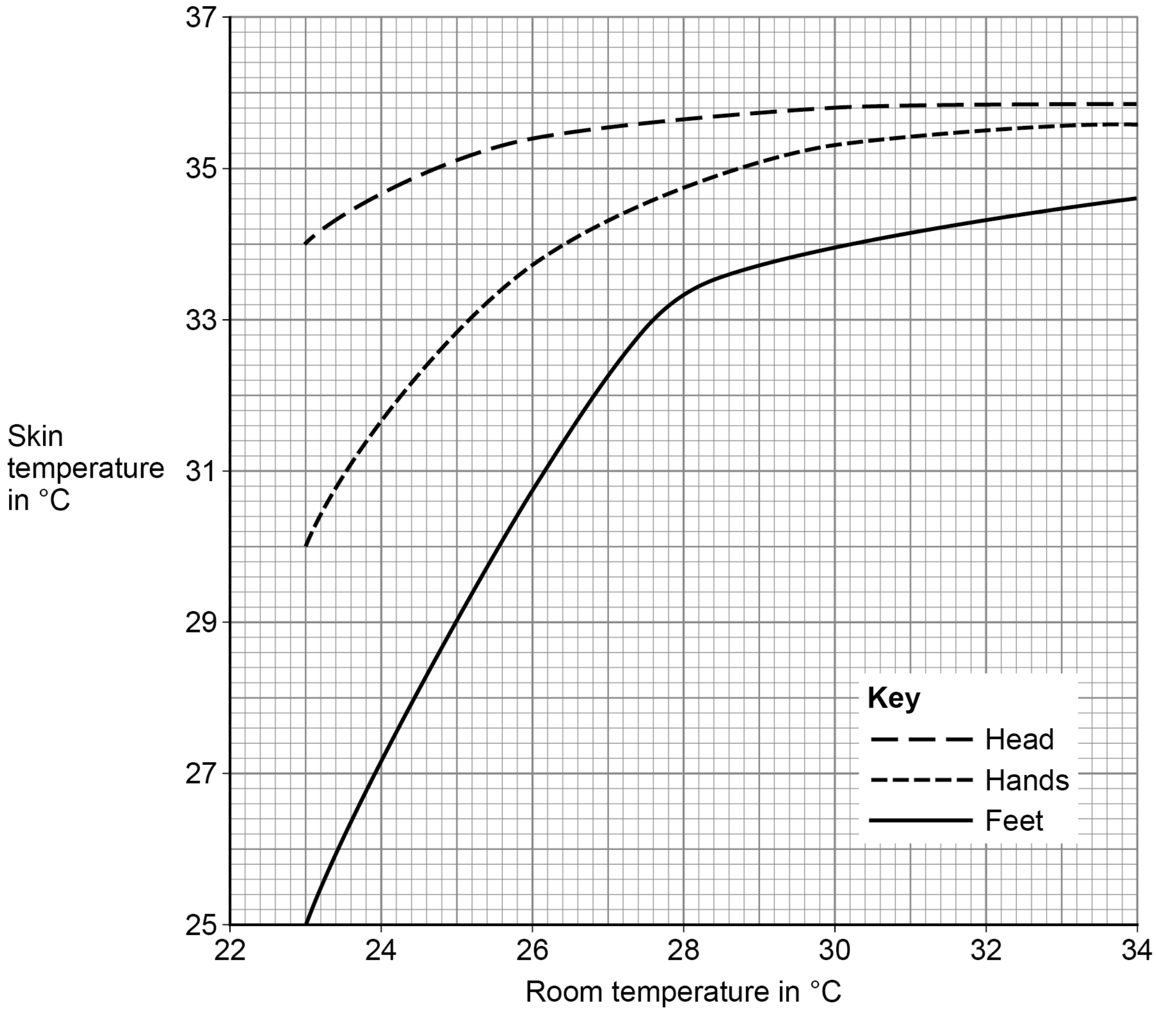
0 5 . 4 An investigation was carried out into the effect of increasing room temperature on skin temperature.

This is the method used.

1. Sit 10 people in a room kept at a specific temperature for 20 minutes.
2. Measure the skin temperatures of different parts of the body.
3. Repeat in rooms at different temperatures.

Figure 8 shows the results.

Figure 8



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Compare the effect of increasing room temperature on the skin temperature of the different parts of the body.

[3 marks]

8

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED

0 6 Bread can be made using anaerobic respiration of yeast.

0 6 . 1 Complete the word equation for anaerobic respiration in yeast.

[2 marks]

Glucose → ____________________ + ____________________

0 6 . 2 A scientist investigated the rate of anaerobic respiration in four species of yeast.

Give **one** condition that is required to make sure the yeasts only respire anaerobically.

[1 mark]

____

____

Question 6 continues on the next page

One of the products of anaerobic respiration in yeast is acidic.

The scientist:

- measured the decrease in pH in 4 hours
- repeated the investigation 20 times and calculated the mean value for each species
- calculated the standard deviation (SD) for the results of each species.

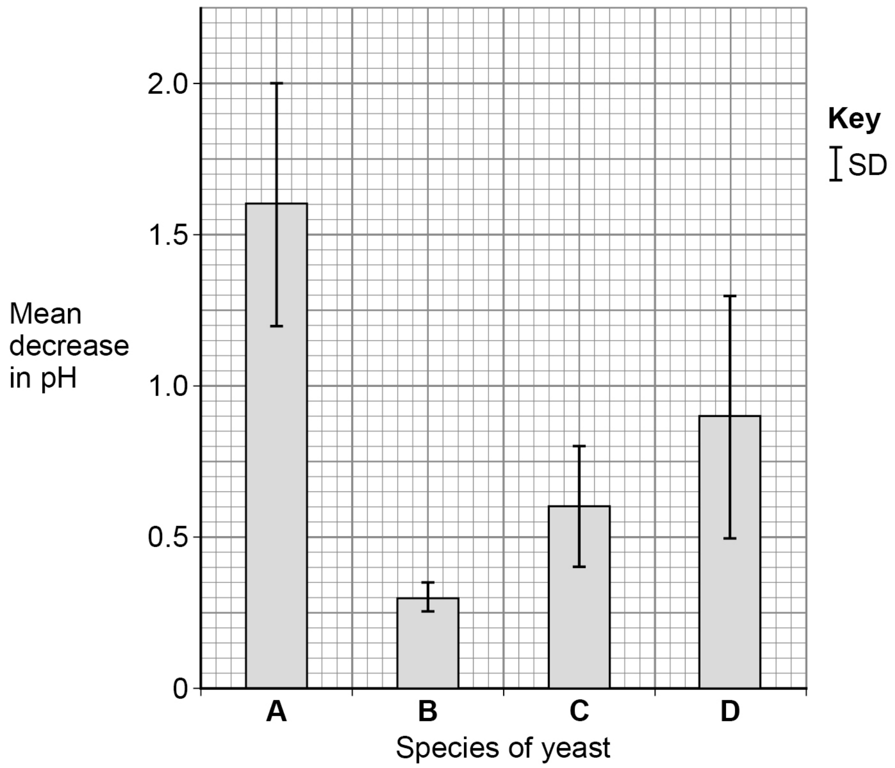
SD is a measure of the spread of the individual results above and below the mean value.

The greater the spread of results around the mean, the greater the SD.

The difference between the means is not significant if there is an overlap in SD.

Figure 9 shows the results.

Figure 9



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0 6 . 3 The scientist stated: 'Species A has the highest rate of anaerobic respiration'.

Evaluate the scientist's statement.

[6 marks]

9

Turn over for the next question

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0 7 Information about inherited characteristics is passed from parent to offspring as a code.

The code is contained on molecules of DNA.

0 7 . 1 Molecules in cells vary in size.

Table 4 shows the lengths of some molecules in cells.

Table 4

Molecule	**Length of molecule**
DNA	220 centimetres
Glucose	0.0015 micrometres
Haemoglobin	5.5 nanometres
Water	0.275 nanometres

Arrange the molecules in **Table 4** in order of length from the shortest to the longest.
[3 marks]

Shortest ____________________

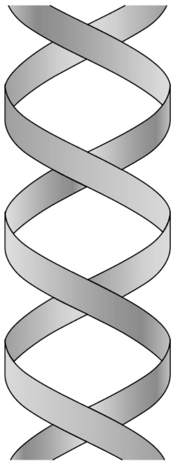
____

____

Longest ____________________

Figure 10 shows part of a DNA molecule.

Figure 10



0 7 . 2 What is the scientific term for the twisted spiral shape shown in **Figure 10**?

[1 mark]

0 7 . 3 Explain how the sequence of bases in a DNA molecule codes for different proteins.

[2 marks]

C2 protein is used to make a strong smooth material that covers the ends of bones in joints.

Syndrome **S** is an inherited disorder that affects joints.

The gene that produces the C2 protein is faulty in a person with syndrome **S**.

0 7 . 4 Suggest **one** symptom people with syndrome **S** may show.

[1 mark]

Turn over ►

In syndrome **S** the code for one amino acid is replaced with a stop code.

A stop code ends the sequence of amino acids.

The short C2 protein that is made does not work.

Table 5 gives some base codes for amino acids needed to make the C2 protein.

Table 5

Base codes	Codes for
CGT or CGG	alanine amino acid
ACG	cysteine amino acid
CCA or CCG	glycine amino acid
AGT or AGC	serine amino acid
ATC or ACT	stop amino acid sequence

Table 6 gives part of the base code for C2 protein in three people, **X**, **Y** and **Z**.

Table 6

Person	Part of their C2 base code
X	C G T A C G C C A A C G C C G A G T C G G A C G C C G
Y	C G T A C G C C G A G T C C G A G T C G G A C G C C A
Z	C G T A C G C C A A C T C C G A G T C G G A C G C C A

0 7 . 5 Identify the person with syndrome **S**.

Give the reason for your answer.

Use **Table 5** and **Table 6**.

[2 marks]

Person ____________

Reason ____

____

Another syndrome, syndrome **K**, is **not** caused by a faulty gene.

In males with syndrome **K** the cells contain two X chromosomes and one Y chromosome (XXY).

Syndrome **K** is caused by an extra chromosome in the nucleus of the fertilised egg cell.

0 7 . 6 The new cell produced at fertilisation has three sex chromosomes, XXY.

Explain how having three chromosomes can be caused by:

- an error in the egg cell
- an error in the sperm cell.

[2 marks]

Error in the egg cell ______________________________

____

Error in the sperm cell ______________________________

____

0 7 . 7 The fertilised egg cell divides many times by mitosis to form the embryo.

Explain how mitosis results in all body cells in the embryo having XXY chromosomes.

[2 marks]

____

____

____

____

____

____

Question 7 continues on the next page

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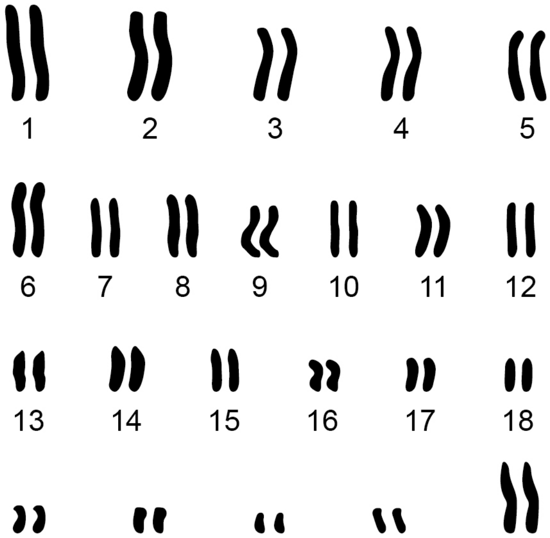
0 7 . 8 **Figure 11** shows a set of chromosomes in a human cell.

Figure 11



Give **two** reasons why this arrangement **cannot** be from a sperm cell.

[2 marks]

1 ____

____

2 ____

____

15

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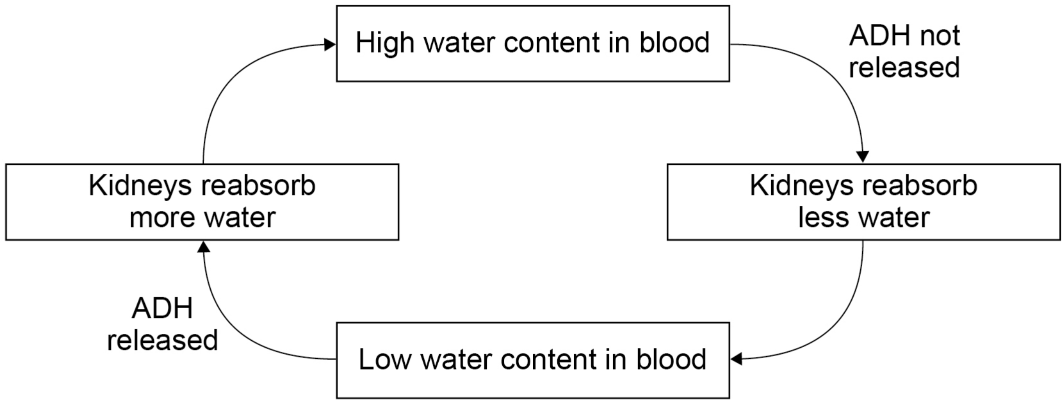
0 8 The hormone ADH controls the water content of the blood.

ADH causes an increase in the amount of water reabsorbed by the kidneys.

Figure 12 shows how ADH controls the water content of the blood.

Figure 12



0 8 . 1 What type of control mechanism is shown in **Figure 12**?

[1 mark]

____

0 8 . 2 Describe the role of the pituitary gland in controlling the water content of the blood.

[2 marks]

____

____

____

____

0 8 . 3 The kidney is the target organ for ADH.

How does ADH reach the kidney?

[1 mark]

____

____

Turn over ►

Diuretic drugs reduce blood volume.

Lower blood volume decreases high blood pressure.

Researchers investigated the effect of different doses of diuretic drug **A**.

10 000 patients with high blood pressure were given a tablet for 8 weeks.

The tablet contained different doses of drug **A**.

Table 7 shows the mean results after 8 weeks.

Table 7

Dose in mg/day	Decrease in blood pressure in arbitrary units
0.0	0
6.5	4
12.5	6
25.0	8
50.0	11
100.0	12

Do not write outside the box

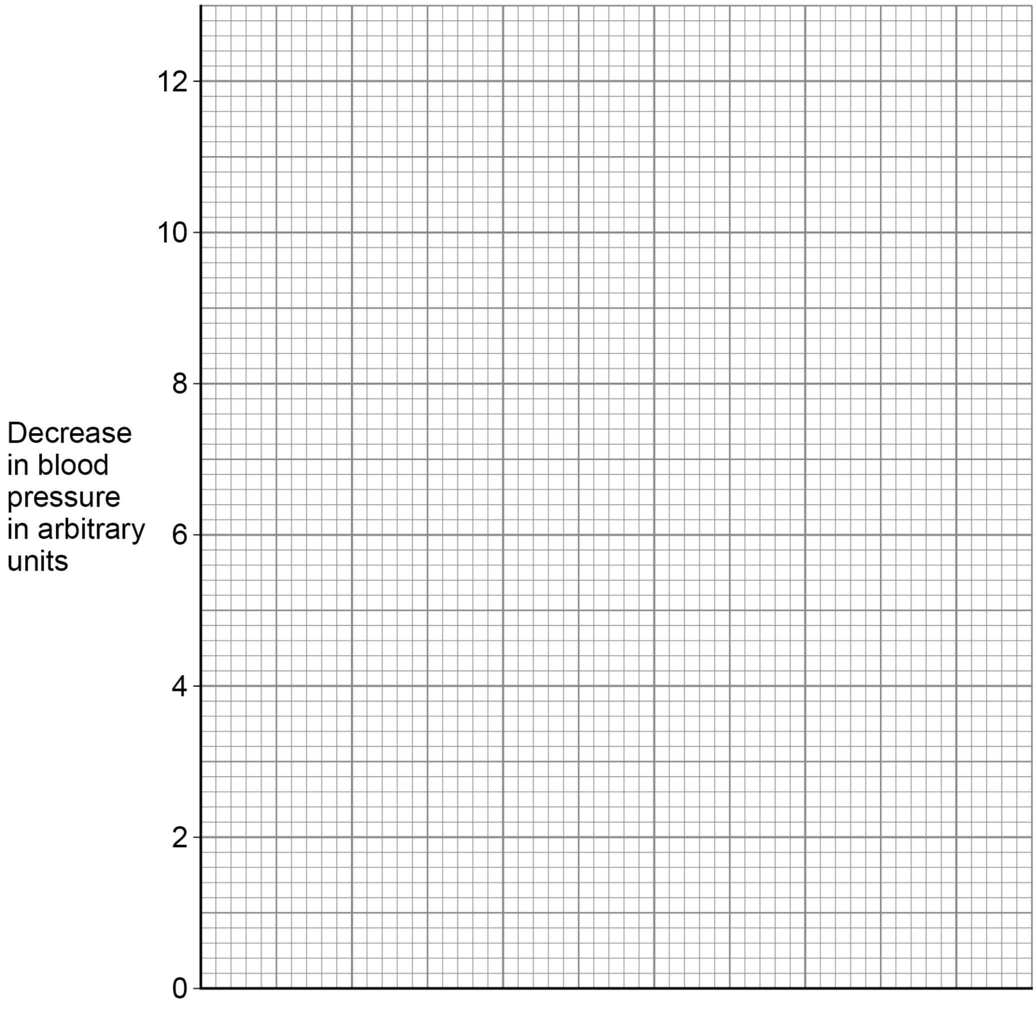
0 8 . 4 Complete **Figure 13**.

You should:

- label the x-axis
- use a suitable scale for the x-axis
- plot the data from **Table 7**
- draw a line of best fit.

[4 marks]

Figure 13



Question 8 continues on the next page

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Table 7 has been repeated here.

Table 7

Dose in mg/day	**Decrease in blood pressure in arbitrary units**
0.0	0
6.5	4
12.5	6
25.0	8
50.0	11
100.0	12

0 8 . 5 Suggest **two** reasons why a dose of 50.0 mg/day of drug **A** was recommended for patients with high blood pressure.

Do **not** refer to cost in your answer.

[2 marks]

1 ____

____

2 ____

____

Do not write outside the box

0 8 . 6 Drug **A** is a diuretic drug that reduces the reabsorption of sodium ions from the kidney.

Explain how drug **A** will cause a decrease in blood pressure.

[4 marks]

14

END OF QUESTIONS

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Question number	**Additional page, if required. Write the question numbers in the left-hand margin.**

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